

15、Infrared remote control

- 1) The infrared remote control receiving range is 8-12 meters. The more open the field is, the smaller the electromagnetic interference is, the farther the remote control distance is.
- 2) The infrared remote controller can only perform on-the-spot remote control on the actuator. Menu setup and data calibration can not be performed.
- 3) Click the reset (RESET) button to remove the remote control state and remove all the "E" errors of the locator.
- 4) Press any other key to start the remote control state.
- 5) In the field remote control operation, attention should be paid to distinguishing broadcast mode from radio mode, so as to avoid the loss caused by errors in remote control operation of field equipment.。



NO	Key	function	illustration
1	RESET	Reset	1. press the button to turn off the remote control (the green light goes out), so that the locator returns to the automatic running state. 2. Note that when an infrared remote control is enabled and then returned to the automatic control state, all "E" errors in the locator are cleared and the automatic control returns to normal control state.
2	SCTR	Single key	3. Independently control the address corresponding to 0-25 (factory address default = 0, can be set by U14, see "485 bus control" address settings method). 4. In the infrared remote control state, click this key, within the range of the received locator, the 0 address locator is valid (the 0 address locator green light is always on), while the other address locators in the remote control receiving range are in the waiting state (the green light flashes). 5. press the key again, and the address is increased by 1, and the 1 address is valid. The address range is 0---25 and cycle. 6. Note that when using this key, keep all localizer signal received in this range reliable, otherwise there may be asynchronous address switching.
3	OPEN/+	Open operation	7. Hold this key, the actuator runs in the open direction (green strobe), stops at 100.0%(green light is always on), releases this key halfway, and the actuator stops running.
4	SHUT/-	Close operation	8. Hold this key, the actuator has been running in the direction of off (green strobe), turn off 0.0% stop (green light is always on), release this key halfway, the actuator stopped running.
5	OPENEND	Fully opened	9. Press this key and the actuator will stop until 100%.
6	SHUTEND	Fully closed	10. Press this key and the actuator will stop until 0%.

16、485 bus control

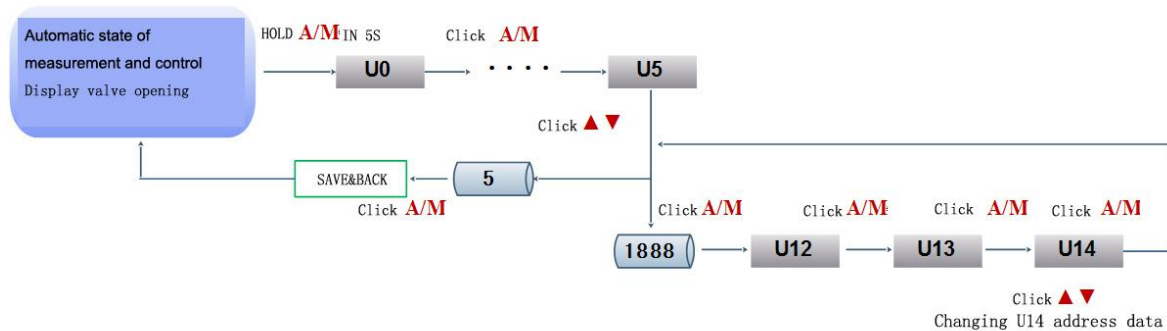
(ONE)、Introduction of 485 bus control:

RS485 communication is widely used in industrial automation control and long-distance data transmission. Usually, data transmission specification adopts MODBUS protocol. In practical applications, 485 relay amplifier is usually added to the subsystem in order to increase the reliability of data transmission, long-distance, anti-interference, and load-carrying capacity. If the transmission distance is less than 300 meters, in 485 load less than 25 systems, no relay (environmental factors

should be considered). The use of shielded cables is a good choice for data transmission lines. When laying lines, we should avoid magnetic field sources, possible external disturbances such as lightning interference.
(TWO)、Additional instructions for locator menu:

Positioner software has added "address" settings. In communication systems, each valve address is unique. In the first installation of the executing agency, the unique address setting is achieved by setting the menu U14 of the locator.

The setting range of false U14 is 0-25, and the maximum corresponds to 26 addresses. The upper computer software must operate corresponding to the actuator valve and must correspond to the corresponding address.



Setting method:

As shown, in the automatic control state, hold down the A/M key, enter the U0 menu, press the A/M key until U5; then press the_key to adjust the data of U5 until 1888, press the A/M key, enter U12, then continue to press U14; at this time according to the address needs, press_key to adjust the data of U14, the data represents positioning. The address of the device. After setting up, press A/M key to return to U5, press_key to adjust the data of U5, make it equal to 5, press A/M key to save exit. The address is set successfully. LCD products, select Advanced Settings > address settings.

Note: view the advanced settings menu table, communication time /U22, factory default is 10. It shows that no communication instructions have been received in 10 seconds, and E1 signal interruption will be reported. This time can be set up to 240 seconds. If you want to block the alarm, set it to 0 and save the exit.

Three)Communication mode:

1. Communication interface electrical standard: Communication interface conforms to RS-232 RS-485 international standard.。

2. Communication protocol: Adopt MODBUS communication protocol RTU mode.。

3. Communication format Convention:

(1) Communication rate: 9600

(2) The character format is 11 bits:

A starting bit, eight data bits, and a parity bit.

A stop bit (check time) and two stop bits (when no verify).

Each character or byte is sent in the following way (from left to right):

Least significant bit... The most significant bit.;

4. RTU frame: Initial structure ≥ 4 byte time Address code = 1 bytes Function code = 1 byte
 Data area = n byte Error check = 16 bit CRC code End structure is more than 4 bytes time.

Address code: the address code is the first byte of the communication. Each slave in a communication network corresponds to a unique address code, which can be set in the range of 0 - 25 (decimal), according to the MODBUS communication protocol, the possible slave device address range is 1 - 25 (decimal). The address code sent by the host indicates the slave address to be sent, and the slave address code sent by the slave indicates the slave address to be sent back.

Function code: the function code is the second byte of the communication. The function definition of MODBUS protocol is 1 - 127. Only one part of the function code is used in this table. The function code sent by the host tells the slave what should be done. Normally, the slave will perform the operation and send back the function code as a response. If the function code sent back has a maximum of 1, it indicates that the slave has not responded to the operation or sent an error.

Data area: the organization of data area is different according to different function codes. The data area can be either the actual value or the address or status setting of the instrument register.

Error check: CRC check. The whole RTU frame is regarded as a continuous binary data string, and the parity check only involves the data bit, leaving out the start bit, stop bit and optional parity check bit.

The steps to generate CRC code are as follows::

- (1) Preset a 16-bit register to be all 1 (hexadecimal FFFFFFFH), which is called CRC register.
 - (2) the first 8 bit data is different from the low 8 bit of the CRC register or the result is returned to the CRC register.
 - (3) shift the 16 bit CRC register to the right one and use 0 to add the highest position to detect the location.
 - (4) Repeat step 3 (move out again) if the offset is 0; if the offset is 1, the CRC register
 - (5) unlike the polynomial A001H, the result is returned to the CRC register.
 - (6) repeat third, fourth steps until the 8 position is removed.
 - (7) Put the next 8-bit data into the CRC register and repeat the second, third, fourth and fifth steps.
- The final CRC register content is the generated CRC check code.

Locator default: communication baud rate 96008 bit data 1 bit stop bit no check; in continuous transmission and reception, send an instruction data needs 10 ms, receive a set of data up to 20 ms. Therefore, for reliable application, >=30ms is recommended to place a single instruction to the locator.

(four) register description:

Instructions for internal registers (double byte for registers and data)

NO	FUNCTION	Read / write Function code	register	Read / write Range of data values	Illustration
1	Bus / simulation	0x06	0x0000	0x0000/0x0001	When the register is written to 0x0001, the locator operates on bus 485.
		0x03		0x0000/0x0001	When the register is written to 0x0000, the locator works on analog control. The register reads the value and reflects the current working state.
2	Valve position value	0x03	0x0001	0xFFFF	The register reads to the value, -1999, and the result is the actual valve position.
3	Set value	0x06	0x0002	0x07CF~0x0BB7	When you write the register data, you need to set the actual setting value +1999.
		0x03		0x07CF~0x0BB7	The register reads the value, -1999, as the actual setting value.
4	Fault code	0x06	0x0003	0x0000	When the register is written to 0x0000, it is clear that all faults are alarms.
		0x03		0x0000~0x000F	The register reads the value, meaning the fault code.
5	Reset / calibration	0x06	0x0004	0x0F30/0x000A	When the register is written to 0x0F30, the locator is reset and reset.
		0x03		0x0000/ 0x0F30/0x000A	When the register is written to 0x000A, the valve position angle / up and down is automatically calibrated. The register reads to the value, 0x0000 (no task). 0x0F30/0x000A (executing the task)

- According to the Modbus specification, the function code 03 is read and 06 is written.
- When the positioner is in the initial state of power-on: when setting U0 = x.2, power-on is 485 communication state. Before receiving the new set value, start / stop / turn-off action is in interrupt mode; when U0 = x.0 or x.1, power-on defaults to analog (such as DC4-20mA) signal control or switching signal control signal is effective.

(five), function example:

1. Read a single register content (function code is 03).:

The host reads the current valve position from the locator and sends it as follows (16 binary byte).:

Send out format	Address	Function code	register High byte	register Low byte	data High byte	data Low byte	CRC check Low byte	CRC check High byte
Send out	01	03	00	01	00	01	D5	CA
(explain)	No. 1 Positioner	read	Valve position register		Read only valve position data		According to the CRC16 specification and method, the check word is calculated and sent to the locator.	

The data returned by the locator contains address, function code, data length, current threshold value and CRC check code. .

Send out format	address	Function code	data length	data High byte	data Low byte	CRC check low byte	CRC check High byte
Give an example	01	03	02	09	C3	FE	45
(explain)	No. 1 Positioner	read	2 byte data	Read only valve position data		Check words according to CRC16 specification and method.	

We check the received data 01 03 02 09 C3 with CRC 16 and get 0xFE45, which is equal to the received CRC check byte FE (low) 45 (high). That is, the returned data is valid and successful.

This example reads out the threshold value of positioner 1: 0x09C3, decimal is 2499; according to the data algorithm: $2499 - 1999 = 500$; because positioner data retains a decimal point, the current valve position is 50.0%.

2. Write a single register content (function code 06).:

The host writes the set value to the locator and sends it as follows (16 binary byte).:

Send out format	address	Function code	register High byte	register Low byte	data High byte	data Low byte	CRC check Low byte	CRC check High byte
Send out	01	06	00	02	0B	B7	6F	4C
(explain)	No. 1 Positioner	write	Set register		Set value data		According to the CRC16 specification and method, the check word is calculated and sent to the locator.	

The data returned by the locator contains address, function code, register + data (or error response value) and CRC check code. .

Send out format	address	Function code	register High byte	register Low byte	data High byte	data Low byte	CRC check Low byte	CRC check High byte
Send out	01	06	00	02	0B	B7	6F	4C
(Explain)	No. 1 Positioner	write	Set register		Set value data		According to the CRC16 specification and method, the check word is calculated and sent to the locator.	

We check the received data 01 06 00 02 0B B7 with CRC 16 and get 0x6F4C. Compared with the received CRC check value byte 6F (low) 4C (high), the CRC check result is equal. That is, the returned data is valid, and the valve positioner successfully receives and executes the instruction.

This example No. 1 positioner successfully set 100.0%, so that the valve position to fully open; data removal decimal point $1000 + 1999 = 2999$, hexadecimal is 0x0BB7; so the register 0x0002 write data 0x0BB7, the control valve fully open 100.0%.

3. Write the contents of a single register - Auto Calibration Valve Position (function code 06): (This function requires actuator limit support)

The main engine reads 0003 registers to determine the valve position blockage failure E4 or E5, and the

valve position needs to be re-calibrated (after calibration, the locator is clear of the fault). Send the following (16 binary byte):

Send out format	address	Function code	register High byte	register Low byte	data High byte	data Low byte	CRC check Low byte	CRC check High byte
Send out	01	06	00	04	00	0A	48	0C
(explain)	No. 1 Positioner	write	Reset / calibration register		Automatic calibration of valve position		According to the CRC16 specification and method, the check word is calculated and sent to the locator.	

After receiving the calibration instructions, the data returned by the locator includes address, function code, register + data (or error response value) and CRC check code. This indicates that the reception is successful and the calibration task is being processed.

Send out format	address	Function code	register High byte	register Low byte	data High byte	data Low byte	CRC check Low byte	CRC check High byte
Send out	01	06	00	04	00	0A	48	0C
(explain)	No. 1 Positioner	write	Reset / calibration register		Automatic calibration of valve position		According to the CRC16 specification and method, the check word is calculated and sent to the locator.	

The locator takes 30-120 seconds (depending on the stroke time of the valve position, and the calibration time is different. For example, 30 seconds actuator, about $30 * 2 + 20$) calibration valve position, during calibration, you can read the value of 0x0004 register to determine whether the automatic calibration valve position is complete.

The host sends the following (16 binary byte):

Send out format	address	Function code	register High byte	register Low byte	data High byte	data Low byte	CRC check Low byte	CRC check High byte
Send out	01	03	00	04	00	01	C5	CB
(explain)	No. Positioner	Read	Reset / calibration register		Read only the register data.		According to the CRC16 specification and method, the check word is calculated and sent to the locator.	

The data returned by the locator contains the address, function code, data length, current 0x0004 register value, and CRC check code.

Send out format	Address	Function code	data length	data High byte	data Low byte	CRC check low byte	CRC check High byte
举例	01	03	02	00	0A	38	43
(explain)	No. 1 Positioner	read	2 byte data	Read only valve position data		Check words according to CRC16 specification and method.	

When the data we read out has been 0x000A, it is still in the calibration of valve position.

When the data read is 0x0000, the calibration is completed. We will write to its normal valve position control instructions again.

After the calibration is completed, the valve will press the last set value action, until the new set value is received, then the new set value action.

4. Read multiple register contents (function code 03):

The host reads the value of register 0x0000~0x0003 from the locator and sends it as follows (16 binary byte):

Send out format	address	Function code	register High byte	register Low byte	data High byte	data Low byte	CRC check Low byte	CRC check High byte
Send out	01	03	00	00	00	04	44	09
(explain)	No. 1 Positioner	read	First register		There are 4 consecutive registers.		According to the CRC16 specification and method, the check word is calculated and sent to the locator.	

The data returned by the locator contains address, function code, data length, locator control status, threshold value, setting value, fault code and CRC check code.。

Send out format	address	Function code	Data length	data High byte format	data Low byte	data High byte	data Low byte	data High byte	data Low byte	data High byte	data Low byte	CRC check low byte	CRC check High byte
Give an example	01	03	08	00	01	0A	4D	0A	4F	00	00	9B	BC
(explain)	No. 1 Positioner	read	2 byte data	Bus state		Valve position data		Setting data		No fault		Check words according to CRC16 specification and method.	

We checked the data of the 1 locator backhaul and checked it correctly. This example reads out the positioner control state: 0x0001 (bus state), the current threshold value 0x0A4D (63.8%), the current set value 0x0A4F (64.0%) and the fault code 0x0000 (fault-free).

(six) response codes:

The function of the response value returned by the locator (16 binary byte):

Response code	Functional description
00	Correctly responds to host operation and sends back the original function code.
01	Illegal function code, and the maximum location of the function code is 1.
02	Illegal registers, and the maximum location of the function code is 1.
03	Illegal data, and function code at the top 1.
04	CRC check error, and the maximum location of the function code is 1.
06	Locator is processing host request, busy (reserved)
12	The locator failed to complete the task assigned by the host.

When 01~04 item is used, the highest position of the function code is 1, which means the 86/83 state.

(seven) fault codes:

Function of response value returned by locator (16 hexadecimal)

Fault code	Functional description
0000	No fault
0002	E1: analog 4-20mA interrupt state of input signal
0004	E2: Potentiometer data do not meet the requirements, check the potentiometer in the valve full close (100-1000) -- full open (2000-4000) data is required; or the potentiometer wire is not connected properly.
0008	E3: valve position data running in the opposite direction, more than 2%
0010	E4: valve turn off.
0020	E5: valve opening direction blocking
0100	E8: locator internal temperature exceeds the set value (factory 85 degrees)

A byte binary 0B00000000 can indicate 8 + 8 01 such as E1 and E4 failures, and the received fault code is: 0x0012